ADVANCED LINEAR ALGEBRA
HOMEWORK 5

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P.3.1 Let $X = [X_1 \quad X_2] \in \mathbf{M}_{m\times n}$ in which $X_1 \in \mathbf{M}_{m\times n_1}$ $X_2 \in \mathbf{M}_{m\times n_2}$, and $n_1 + n_2 = n$. Compute $X^\top X$ and $XX^\top$.

Solution. We begin by computing $XX^\top$:

$$XX^\top = \begin{bmatrix} X_1 & X_2 \end{bmatrix} \begin{bmatrix} X_1^\top \\ X_2^\top \end{bmatrix} = X_1X_1^\top + X_2X_2^\top.$$

Now compute the opposite order $X^\top X$ as

$$\begin{aligned} X^\top X &= \begin{bmatrix} X_1^\top \\ X_2^\top \end{bmatrix} \begin{bmatrix} X_1 & X_2 \end{bmatrix} \\ &= \begin{bmatrix} X_1^\top X_1 & X_1^\top X_2 \\ X_2^\top X_1 & X_2^\top X_2 \end{bmatrix}. \end{aligned}$$

□

P.3.12 Compute

$$\det \begin{bmatrix} 0 & 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 0 & 2 & 5 \\ 0 & 0 & 0 & 0 & 0 & 3 \\ 3 & 5 & 6 & 0 & 0 & 0 \\ 0 & 2 & 4 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Solution. Before computing the determinant we do some row permutations. Each time we do a row permutation we have to multiply our determinant by -1. Let r_i represent a row. Our goal is to take our matrix and make it upper triangular, so that all we need do is multiply the values on the main diagonal together to compute the determinant. The row permutations

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follow as $r_1 \leftrightarrow r_4$, $r_2 \leftrightarrow r_5$, and $r_3 \leftrightarrow r_6$. Thus, we get

$$\det \begin{bmatrix} 0 & 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 0 & 2 & 5 \\ 0 & 0 & 0 & 0 & 0 & 3 \\ 3 & 5 & 6 & 0 & 0 & 0 \\ 0 & 2 & 4 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} = -\det \begin{bmatrix} 3 & 5 & 6 & 0 & 0 & 0 \\ 0 & 2 & 4 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 0 & 2 & 5 \\ 0 & 0 & 0 & 0 & 0 & 3 \end{bmatrix} = -36$$

□

P.3.26 Let $A, B \in \mathbf{M}_{m\times n}$ and suppose that $r = \operatorname{rank}(A) \geq 1$. Show that the following are equivalent:

(a) $\operatorname{col}(A) = \operatorname{col}(B)$.
(b) There are full rank matrices $X \in \mathbf{M}_{m\times r}$ and $Y, Z \in \mathbf{M}_{r\times n}$ such that $A = XY$ and $B = XZ$.
(c) There is an invertible $S \in \mathbf{M}_n$ such that $B = AS$.

Hint: If $A = XY$ is a full-rank factorization, then $A = [X \quad 0_{n\times(n-r)}]W$, in which $W = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} \in \mathbf{M}_n$ is invertible.

Proof. We will proceed by showing (b) $\Rightarrow$ (c) $\Rightarrow$ (a) $\Rightarrow$ (b) which by circular implication completes the equivalence of the three statements.

(b) $\Rightarrow$ (c): Let $A = XY$ be a full rank factorization. Then,

$$A = \begin{bmatrix} X & 0_{n\times(n-r)} \end{bmatrix} \begin{bmatrix} Y \\ Y_2 \end{bmatrix} \qquad \text{with} \qquad \begin{bmatrix} Y \\ Y_2 \end{bmatrix} = W \in \mathbf{M}_n.$$

And, W is invertible. But, then $n - r = 0$ and we see that $X, Y \in \mathbf{M}_n$ with Y invertible. We also have $B = XZ$ as a full rank factorization, so by a similar argument Z is invertible. Now substitute in $X = AY^{-1}$ into our B equation and we get

$$B = AY^{-1}Z,$$

and $Y^{-1}Z$ is invertible because both Y and Z are invertible. So if we let $S = Y^{-1}Z$ from our (c) statement, we are done.

(c) $\Rightarrow$ (a): Assume $B = AS$ with $S \in \mathbf{M}_n$ invertible. We know that $\operatorname{col}(AS) \subseteq \operatorname{col}(A)$. So, $\operatorname{col}(B) \subseteq \operatorname{col}(A)$. Similarly, since S is invertible $A = BS^{-1}$, so $\operatorname{col}(A) \subseteq \operatorname{col}(B)$. Now because $\operatorname{col}(B) \subseteq \operatorname{col}(A)$ and $\operatorname{col}(A) \subseteq \operatorname{col}(B)$, we see that $\operatorname{col}(A) = \operatorname{col}(B)$, and we are done.

(a) $\Rightarrow$ (b): Assume that $\operatorname{col}(A) = \operatorname{col}(B)$, and let $A = \begin{bmatrix} X & P \end{bmatrix}$ with columns of X being a basis for $\operatorname{col}(A)$. Notice $\operatorname{col}(X) = \operatorname{col}(A)$ woth dimension r. Each column of A is a linear combination of columns of X, so we can write $A = XY$ for some $Y \in \mathbf{M}_{r\times n}$. Now our goal is to find a $Z \in \mathbf{M}_{r\times n}$ for B similarly so that $B = XZ$. Since $\text{rank}(A) = r$ and $A = XY$ we have

$$r = \text{rank}(XY) \leq \text{rank}(Y) \leq r \quad \Rightarrow \quad \text{rank}(Y) = r.$$

Which is full row rank for XY. Similarly $\text{rank}(B) = r$ by $\text{col}(A) = \text{col}(B)$ which gives us

$$r = \text{rank}(XZ) \leq \text{rank}(Z) \leq r \quad \Rightarrow \quad \text{rank}(Z) = r.$$

So $B = XZ$ has full row rank, and we are done.

Lastly we notice that (b) $\Rightarrow$ (c) $\Rightarrow$ (a) $\Rightarrow$ (b) so our equivalence holds. □

P.3.27 Let $A, C \in \mathbf{M}_m$ and $B, D \in \mathbf{M}_n$. Show that there is an invertible $Z \in \mathbf{M}_{m+n}$ such that $A \oplus B = (C \oplus D)Z$ if and only if there are invertible $X \in \mathbf{M}_m$ and $Y \in \mathbf{M}_n$ such that $A = CX$ and $B = DY$.

Proof. We begin by noting that we can write the following

$$A \oplus B = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \quad \text{and} \quad C \oplus D = \begin{bmatrix} C & 0 \\ 0 & D \end{bmatrix}$$

($\Leftarrow$) Assume that we have Z invertible in the following fashion

$$Z = X \oplus Y = \begin{bmatrix} X & 0 \\ 0 & Y \end{bmatrix}.$$

We then expand in the following fashion

$$(C \oplus D)Z = \begin{bmatrix} C & 0 \\ 0 & D \end{bmatrix} \begin{bmatrix} X & 0 \\ 0 & Y \end{bmatrix} = \begin{bmatrix} CX & 0 \\ 0 & DY \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} = A \oplus B.$$

($\Rightarrow$) Assume $(C \oplus D)Z = A \oplus B$ with Z invertible as stated in the problem statement. Notice that we can write

$$Z = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}.$$

with $Z_{11} \in \mathbf{M}_m$, $Z_{22} \in \mathbf{M}_n$, $Z_{12} \in \mathbf{M}_{m\times n}$, $Z_{21} \in \mathbf{M}_{n\times m}$. So we can expand in the following manner

$$(C \oplus D)Z = \begin{bmatrix} C & 0 \\ 0 & D \end{bmatrix} \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} CZ_{11} & CZ_{12} \\ DZ_{21} & DZ_{22} \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} = A \oplus B.$$

So we quickly can see that $Z_{12} = 0$ and $Z_{21} = 0$. Furthermore, we can see that $A = CZ_{11}$ and $B = DZ_{22}$. Because we have that Z is block diagonal and invertible, we have that Z_{11} and Z_{22} are invertible. So with regards to the problem statement we can set $Z_{11} = X$ and $Z_{22} = Y$, and we are done. □

P.3.29 Let $A, B \in \mathbf{M}_{m\times n}$ and suppose there is a $X \in \mathbf{M}_n$ such that $A = BX$. If $\text{rank}(A) = n$ show that $\text{rank}(B) = n$ and X is invertible. What can you say about the converse?

Proof. Assume $A, B \in \mathbf{M}_{m\times n}$, $X \in \mathbf{M}_n$, and $\text{rank}(A) = n$. Write

$$A = \begin{bmatrix} C \\ P \end{bmatrix} \quad \text{where} \quad C \in \mathbf{M}_n,$$

with $P \in \mathbf{M}_{(m-n)\times n}$. But because $\text{rank}(A) = n$, $m - n = 0$ or $m = n$. So now we have that $BX \in \mathbf{M}_n$, and we know that $X \in \mathbf{M}_n$ by our assumptions, so $B \in \mathbf{M}_n$. Now we have that $A = BX$ is a full rank factorization of A, so X must be invertible by problem 3.26. Furthermore, we can write $B = AX^{-1}$ with $AX^{-1} \in \mathbf{M}_n$, so $B \in \mathbf{M}_n$. Given that we have $\text{rank}(A) = n$, and $\text{rank}(X^{-1}) = n$ (by invertibility) we must have $\text{rank}(B) = n$ and we are done.

Regarding the converse it's true, but we leave it as an exercise for the reader :-p (note, it's an analogous proof to 3.26). $\square$